

Xiuyuan Yu

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RESEARCH INTERESTS

Spatial Intelligence · 3D/4D Vision · Representation Learning · World Models · Multimodal Reasoning

EDUCATION

The Chinese University of Hong Kong, Hong Kong SAR

Aug 2023 — Present

B.Eng. in Artificial Intelligence: Systems and Technologies, ELITE Stream

- CGPA: 3.851/4.0, MGPA: 3.963/4.0
- Outgoing Exchange Student
 - University of Illinois Urbana-Champaign, USA, Aug 2025 — Dec 2025
 - Awarded Yasumoto International Exchange Scholarship
- Relevant courses: Digital Signal Processing, Applied Machine Learning, Image Processing and Visual Understanding, Multimedia Coding and Processing, Computer Graphics

SKILLS

Programming Languages: Python (Proficient), C/C++ (Familiar), CUDA (Basic)

Deep Learning Frameworks: Pytorch (Proficient), JAX/Tensorflow (Basics)

Computer Vision & 3D Libraries: OpenCV, COLMAP, Open3D, Kaolin

Development & Tools: Git, Linux, Latex, Docker

HONORS AND AWARDS

Best Project Award for UG Summer Research Internship, <i>Faculty of Engineering</i>	2026
Silver Award for Outstanding Academic Performance (Ranked 1st in major), <i>Department</i>	2025, 2026
Dean's List, <i>Faculty of Engineering</i>	2024, 2025
ELITE Stream Scholarship, <i>Faculty of Engineering</i>	2024, 2025
Yasumoto International Exchange Scholarship, <i>Universtiy</i>	2025
Honors at Entrance (Ranked 329 th in Gaokao, Guangdong), <i>University</i>	2023

RESEARCH EXPERIENCE

Research Assistant & Research Intern.

Hong Kong

MMLab, The Chinese University of Hong Kong

Apr 2025 — Aug 2026

- Advisor: Prof. Tianfan Xue
- Topic: 3D/4D reconstruction and video generation
- Creating the training dataset for a diffusion-based refinement model by reconstructing dynamic scenes with 4D Gaussian Splatting (4DGS).
- Adapt diffusion model for 3D Gaussian Splatting super resolution, enabling close-up novel view synthesis.

PUBLICATIONS

- Chen, Y., Guo, S., Yang, T., Ding, L., **Yu, X.**, Gu, J., & Xue, T. (2025). 4DSloMo: 4D Reconstruction for High Speed Scene with Asynchronous Capture. *ACM SIGGRAPH Asia*.